

(a)



A 35-year-old patient has upper-limb pain. What's the most likely diagnosis?

Where is the pain worst: **forearm or elbow, wrist, or hand**?

Hand.

Do the **fingers change color** when exposed to cold?

No.

When the finger bends or straightens, is there **clicking or catching with pain**?

Yes.

This pattern is most consistent with **trigger finger**.

(b)



Query

Response

Dialogue

Answer

Degree of Underspecification

Instruction (predict k)

How many variables must be asked, at minimum, to uniquely determine Y ?

A.0 B. $k=1$ C. $k=2$ D. $k=3$ E. $k=4$ F.Unsure

Metric

- Accuracy: $1[\hat{k} = k]$
- Under-estimation: $1[\hat{k} < k]$
- Over-estimation: $1[\hat{k} > k]$

Missing Variable Identification

Instruction (select suff. variables)

Select exactly k variables from candidates that are sufficient to determine Y .

Metric

- Exact-match Accuracy: $1[\hat{S} = S^*]$
- Jaccard Similarity
- Recall

Sequential Task-Solving

Interaction (multi-turn)

Turn 1: Ask $X_2 \rightarrow$ Oracle: x_2

Turn 2: Ask $X_3 \rightarrow$ Oracle: x_3

...

Turn t : Stop. Predict $\hat{y}$

Metric

- Final Sufficiency & Answer Accuracy
- Query Coverage, Order Correctness
- Question Quality

(c)

Constraint Satisfaction Problem

Constraints (C) Observed assignment ($\mathcal{A}$) Target variable (Y) Min. sufficient set (S)

k -Underspecified

GSME-Q-MT	Logic-Q-MT	GeneReg-MT	ClinGuide-MT	20Q
$TF = 3^*C$ $TW = 6^*C$ $F3 = TF - F1 - F2$ $W3 = TW - W1 - W2$	$smart \Rightarrow jittery$ $jittery \wedge worried$ $\wedge stubborn$ $\Rightarrow pleasant$ $pleasant \Rightarrow worried$ $pleasant \Rightarrow stubborn$	$A = T1 \text{ and } B \text{ and } H$ $S = H \text{ or } S$ $H = (T1 \text{ and } S) \text{ or } H$ $B = (T1 \text{ or } B) \text{ and } H$...	$worst \text{ pain} = hand$ $\rightarrow Raynaud's \text{ disorder}$ $\rightarrow ask \text{ clicking pain}$ $worst \text{ pain} = forearm$ $\rightarrow epicondylitis$ $\rightarrow ask \text{ erythema}$	$world\text{-}knowledge \text{ constraints}$
$F2 = 25$ $W1 = 10$ $W2 = 40$	$smart = True$	$T1_{init} = 0$	$worst \text{ pain} = hand$	$\emptyset$
$F3 = ?$	$pleasant = ?$	$TO_{steady} = ?$	$diagnosis = ?$	$object = ?$
$F1 = ?$ $C = ?$	$worried = ?$ $stubborn = ?$	$TO_{init} = ?$ $T2_{init} = ?$	$target\text{-}dependent \text{ sufficient queries}$	$candidate \text{ partitions}$

Target	Target	Target	Target	Target
$F3 = 25$	$pleasant = True$	$TO_{steady} = on$	$Raynaud's \text{ disorder}$	$object = rabbit$
Query Set	Query Set	Query Set	Query Set	Query Set
$F1, C$	$worried, stubborn$	$TO_{init}, T2_{init}$	$finger \text{ color change?}$ $clicking \text{ pain?}$	$living \text{ thing?}$ $mammal? \dots$

Task Instance